

- This exam contains 11 pages (including this cover page) and 14 problems. Check to see if any pages are missing.
- This is the third exam and will count for 20% of your final grade. You have 120 minutes (5:00 pm - 7:00 pm) to complete this exam. You may use one index card of notes and a calculator on this exam, but no other outside sources may be consulted.

The following rules apply:

- If a question is followed by work, then work of some form is required. An incorrect answer supported by substantially correct calculations and explanations will receive partial credit. **A correct answer that has no supporting work when work is required will not receive full credit.**
- Organize your work, in a reasonably neat and coherent way, in the space provided. Work scattered all over the page without a clear ordering will receive very little to no credit.
- There are 300 points total in this exam, as well as 2 bonus questions worth a total of 40 possible points.

Page:	3	4	5	6	7	8	9	10	Total
Points:	45	45	30	30	40	40	45	25	300
Score:									

Bonus Points:

1. 25 points Let $f(x) = x^3 - 2x^2 + x + 1$. Find the absolute minimum and the absolute maximum of $f(x)$ over the interval $\left[\frac{1}{4}, 1\right]$, if such an absolute maximum or minimum exists. For every absolute extremum, give the point $(x, f(x))$ and indicate whether it is a minimum or maximum.

work

2. 20 points Let $f(x) = (x - 1)e^x + 2$. Find the absolute minimum and the absolute maximum of $f(x)$ over the interval $(0, \infty)$, if such an absolute maximum or minimum exists. For every absolute extremum, give the point $(x, f(x))$ and indicate whether it is a minimum or maximum.

work

3. 25 points Let $f(x) = (x - 1)e^x + 2$. Find the absolute minimum and the absolute maximum of $f(x)$ over the interval $(-1, 1)$, if such an absolute maximum or minimum exists. For every absolute extremum, give the point $(x, f(x))$ and indicate whether it is a minimum or maximum.

work

4. 20 points A book store in a dark corner of Wyoming sells 40 copies per week of an infamous human-hunting instructional manual titled *The Most Dangerous Game: Advanced Mantrapping Techniques* (yes, this is actually a real book that someone wrote). The books are currently sold for \$60 a copy. A market survey shows that for every \$2 reduction in price, 3 more books will be sold. How much should the book store charge in order to maximize revenue?

work

5. 20 points A box is to be made at a manufacturing company out of a square piece of cardboard 14 inches wide and 14 inches long. Squares of equal size will be cut out of each corner, and then the ends and sides will be folded up to form a rectangular box with an open top. What size square should be cut from each corner to maximize volume?

work

6. 10 points Find $\int \ln(x) \, dx$
(Hint: Try $u = x\ln(x) - x$ for your substitution)

work

7. 15 points Suppose that the marginal profit of producing x beach towels featuring life-sized characters from popular anime programs is given by $P'(x) = e^x - \frac{2}{x^2}$ (where $P(x)$ is the profit in dollars). Suppose we also know that $P(10) = \frac{203}{5}$. Find $P(x)$.

work

8. 15 points Find $\int \frac{x^6 e^x - 3(e^x)^2}{e^x} dx$.

work

9. In each problem, find $\int f(x) \, dx$.

(a)

20 points
work

 $f(x) = (3x + 2)(x^3 + 2x + 1)^{11}$

(b)

20 points
work

 $f(x) = (10x + 2)e^{5x^2+2x+3}$

(c)

20 points

 $x^{15}e^{x^{16}-92}$

work

(d)

20 points

 $f(x) = \frac{24x^{11}}{(x^{12} - 3)^6}$

work

10. 20 points Calculate $\int_2^9 3x^2 + x^3 + 2 \, dx$ given that

$$\int_2^9 x^2 \, dx = \frac{721}{3}, \quad \int_2^{13} x^3 \, dx = 7,136.25, \quad \int_9^{13} x^3 \, dx = 5,500, \quad \text{and} \quad \int_2^9 6 \, dx = 42.$$

work

11. 25 points Let $f(x) = x^2$. Over the interval $[1, 3]$, calculate R_4 , the right-hand Riemann sum.
(Hint: Approximate the area under the graph of f on the interval $[1, 3]$ with boxes of equal width whose top, right-hand corners touch the graph of f)

work

12. 25 points Calculate $\int_{-1}^3 3x^2 + x + 2 \, dx$.

work

13. 20 points (bonus) Find $\int (6x - 1)^9(2x + 4)(3 - x) \, dx$.

work

14. 20 points (bonus) Calculate $\int_0^2 15x^4(x^5 + 2)^{-3} \, dx$.

work